

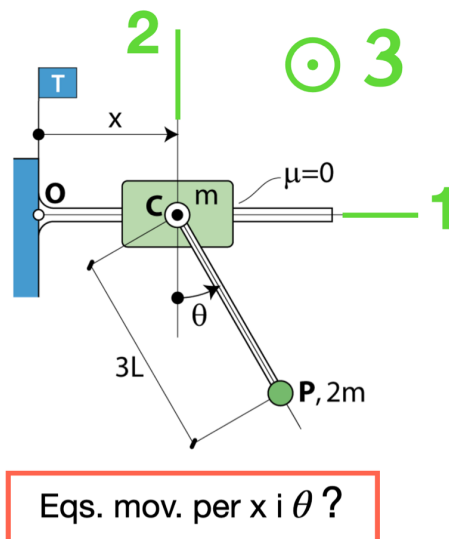
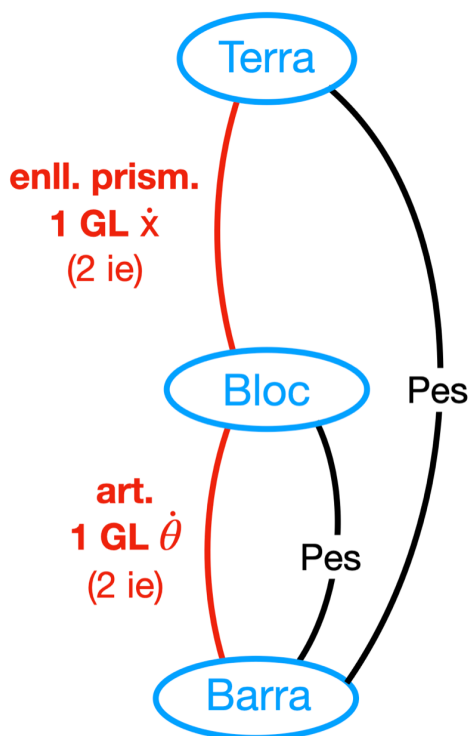
12P PROF

Teoremes vectorials III

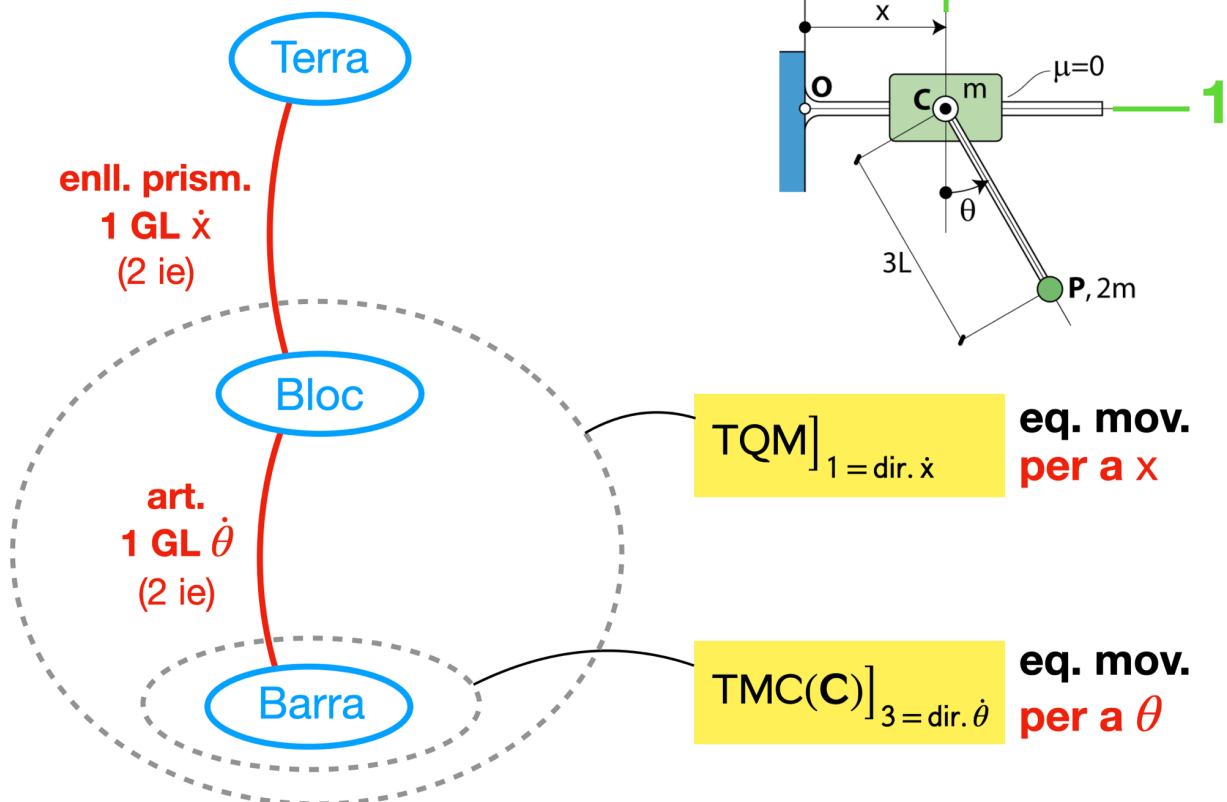
Lluís Ros
<https://lluisros.github.io/mecanica>

2025-26 Q2

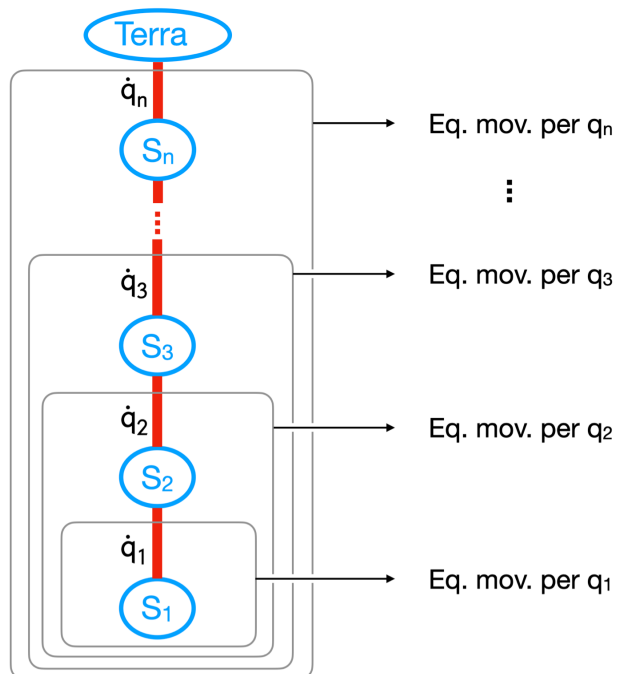
Pèndol d'Euler (recordem)



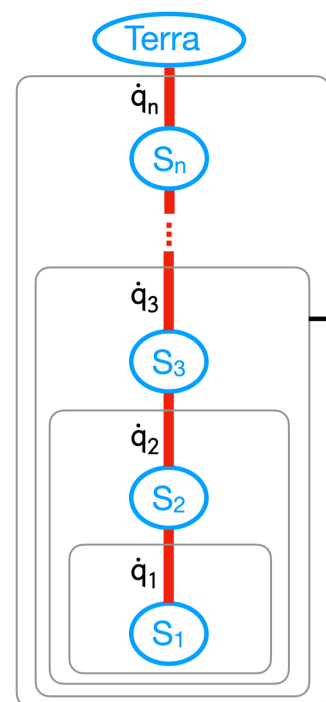
DGIE



En general



En general



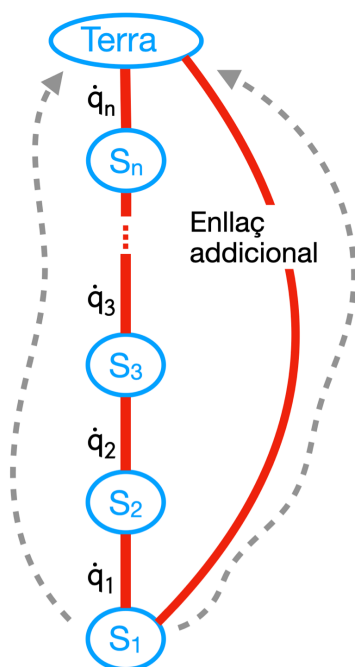
I si algun GL és forçat?

Si un motor garanteix $\dot{q}_3 = ct$, el sistema

$$S_1 + S_2 + S_3$$

no dóna una eq. del mov, sinó l'**acció motora** (força o parell) que assegura

$$\dot{q}_3 = ct$$



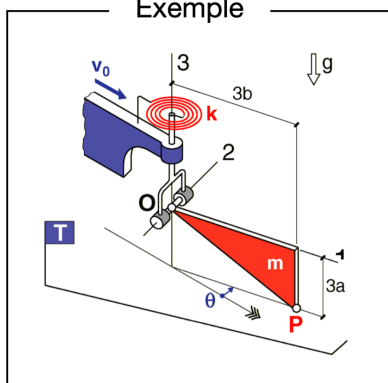
Compte!

Camins múltiples entre S_1 i Terra



No podem aplicar els coneixements previs

Exemple



Pèndol anular giratori

(adaptat de P2, juliol 2016)

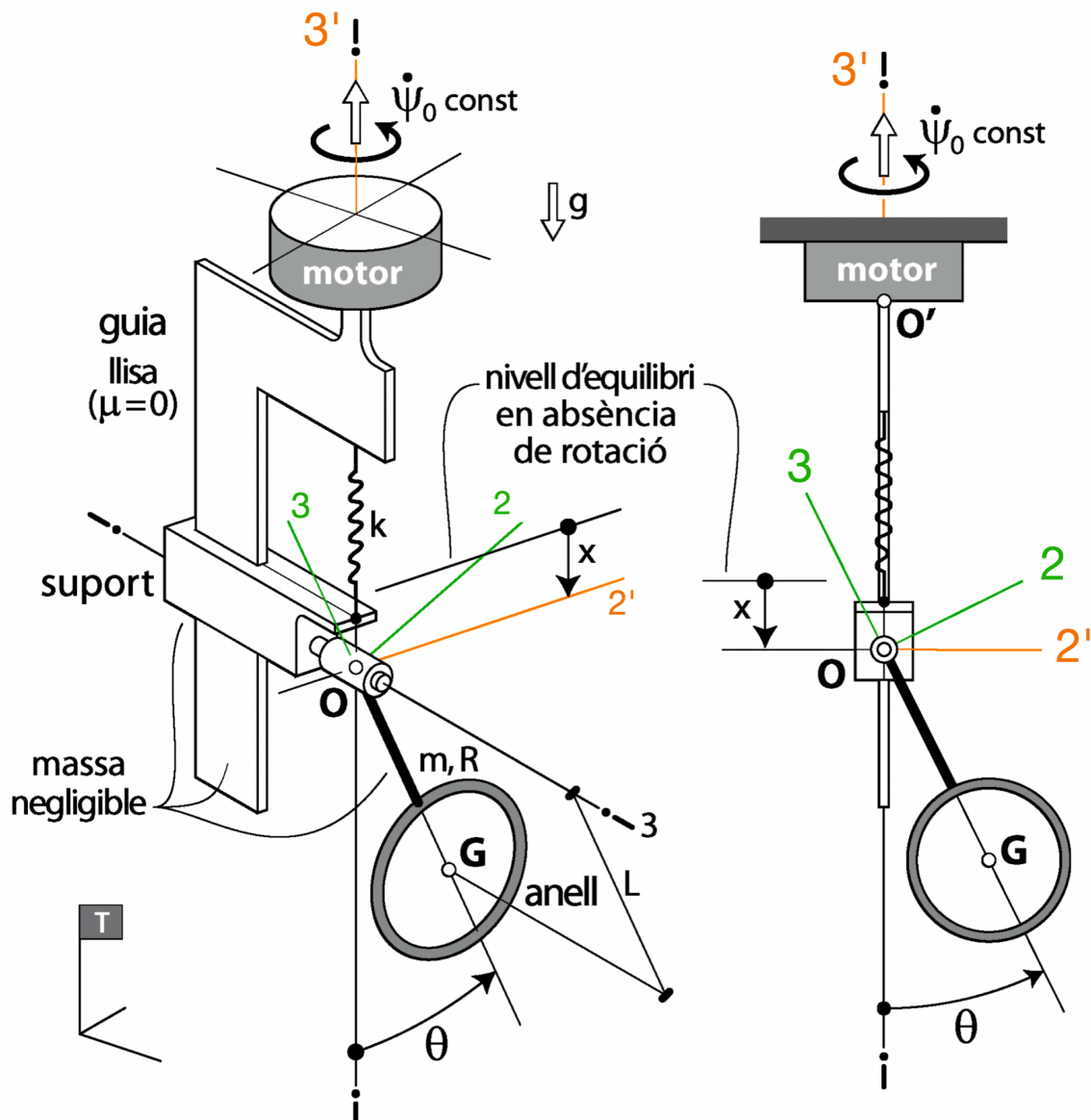
El pèndol, format per un anell homogeni de massa m i radi R i una barra de longitud $(L-R)$, està articulat al punt O del suport, el qual llisca dins d'una guia llisa de secció rectangular. Entre suport i guia hi ha una molla lineal de constant k . Tot el conjunt es mou amb velocitat angular constant $\dot{\psi}_0$ respecte al terra sota l'acció d'un motor.

Amb el motor aturat, la configuració ($x=0$, $\theta=0$) és d'equilibri.

Les masses de la barra, el suport i la guia, i les friccions associades a les articulacions són negligibles.

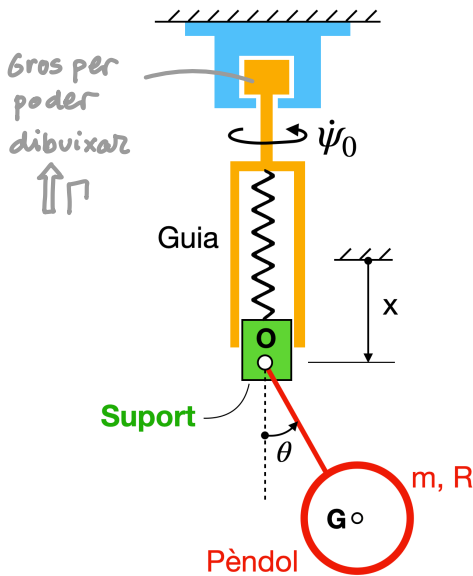
Determineu:

- Les equacions del moviment per a les coordenades x i θ .
- El parell Γ que ha d'aplicar el motor per tal de mantenir $\dot{\psi}_0$ constant.



Eqs. mov. per a x , i per a θ

Dibuix equivalent



El dibuix a dalt dreta
pissarra per tenir-lo a
mà tota l'estona,
mantenint el dibuix
3D realista a pantalla

GL Sistema Ho escric a l'eq. de dibuix:

$$3 \text{ GL} \left\{ \begin{array}{l} \dot{\psi} = \dot{\psi}_0, \text{ forçat} \Rightarrow \text{Parell motor} \Rightarrow \Gamma \text{ és incòg. assoc.} \\ \dot{x}, \text{ lliure} \Rightarrow \ddot{x} \text{ " " "} \\ \dot{\theta}, \text{ lliure} \Rightarrow \ddot{\theta} \text{ " " "} \end{array} \right.$$

3 eqs mov.

- 1 trivial: $\dot{\psi} = \dot{\psi}_0 = ct \Rightarrow \boxed{\ddot{\psi} = 0}$

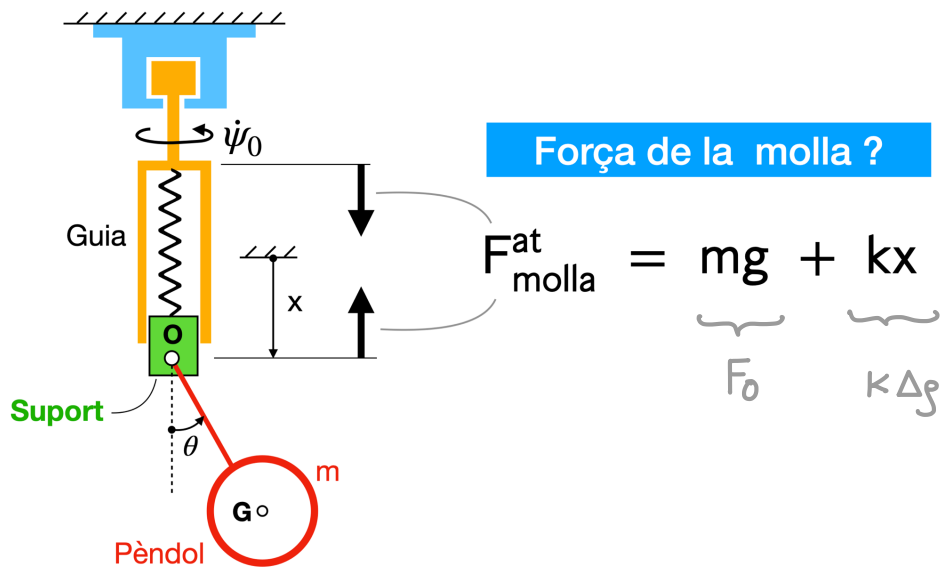
- Les altres dues, de la forma:

$$\begin{array}{l} f_1(x, \dot{x}, \ddot{x}, \theta, \dot{\theta}, \ddot{\theta}) = 0 \\ f_2(x, \dot{x}, \ddot{x}, \theta, \dot{\theta}, \ddot{\theta}) = 0 \end{array} \quad \left| \begin{array}{l} \text{forma} \\ \text{implícita} \end{array} \right.$$

Caldrà buscar 2 eqs. lliures d'ie !

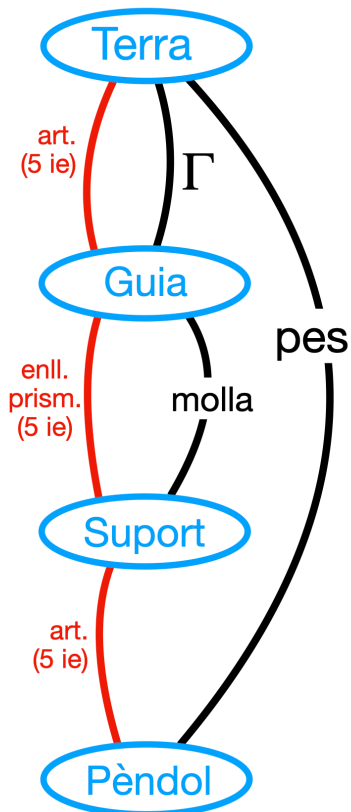
Força molla

La formulem per ús posterior (no entro en gaires detalls)

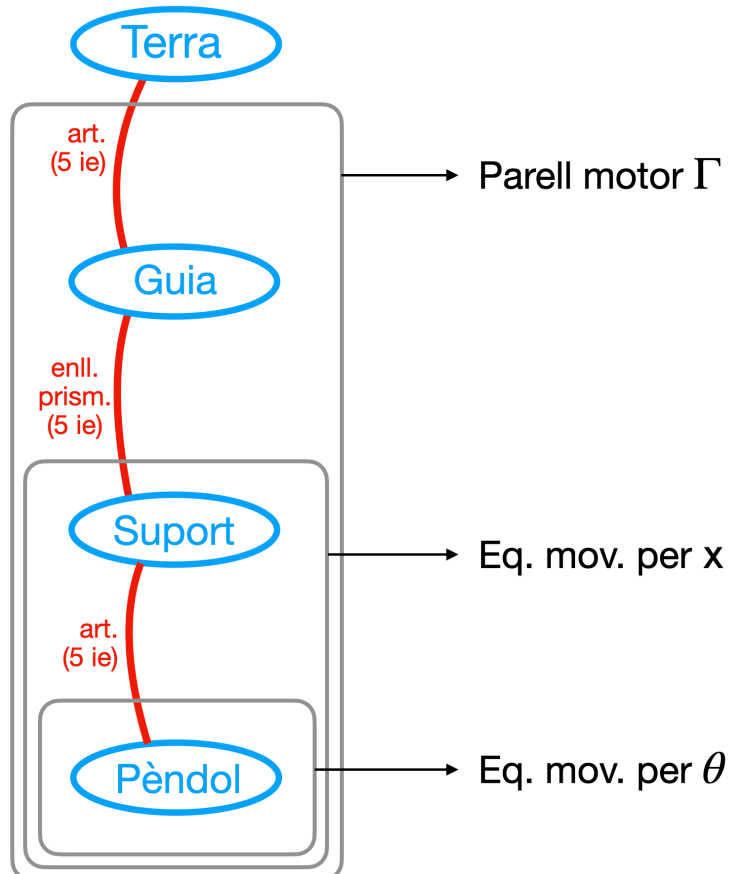


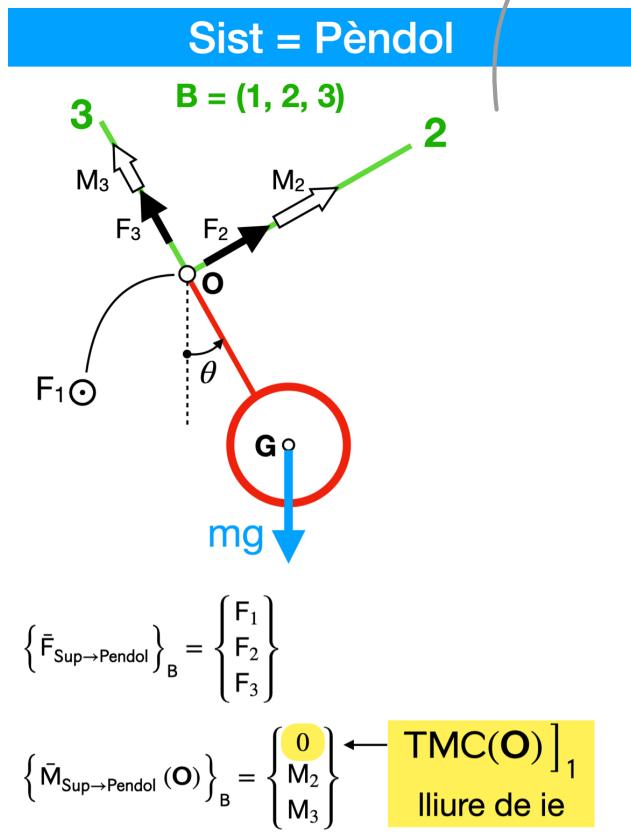
Amb motor aturat: $x = 0$, $\theta = 0$ és configuració d'equilibri.

DGI

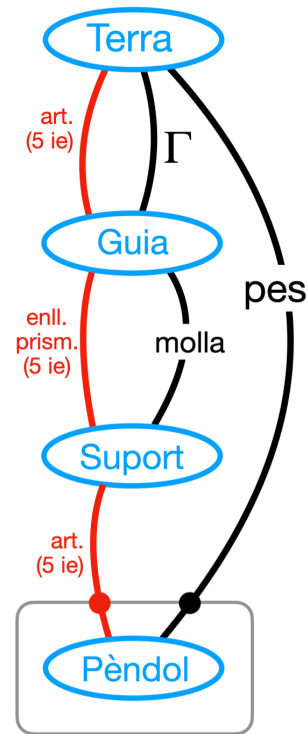


DGIE

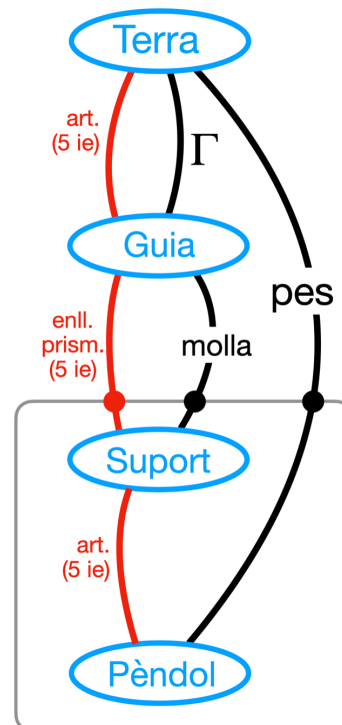
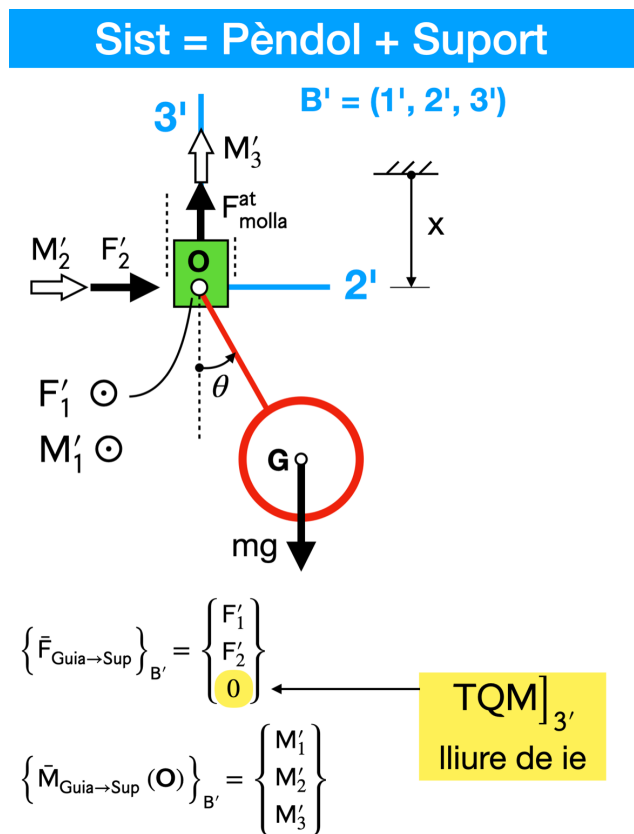




Trio B perquè $\left\{ \begin{array}{l} \text{caract. immediata } \underline{\text{torsor}} \text{ enllaç} \\ \text{tensor inèrcia ct.} \end{array} \right.$



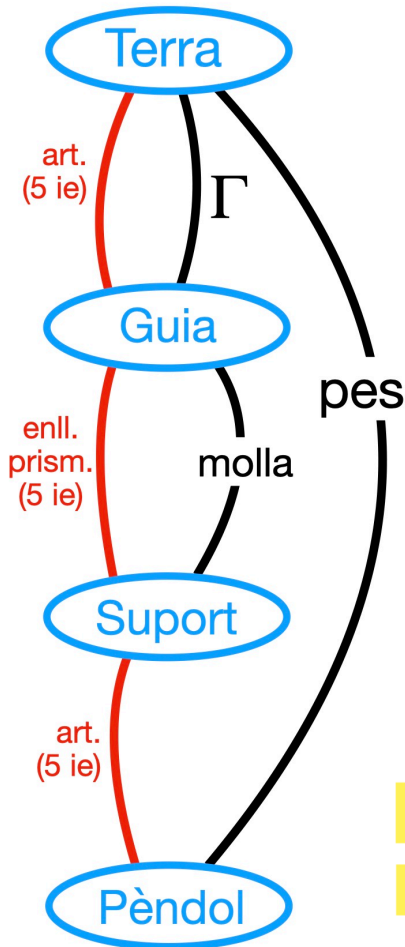
Ergo: Full nota eq. mov. θ Sist = Pèndol, $\text{TMC}(O)]_1$



Ergo: Full nota eq. mov. x SIST = Pènd + Sup, $\text{TQM}]_{3'}$

Els poso junts a pissarra

Ja no ho explico. Amb la nova intro ja no cal !



15 ie, Γ , $\ddot{x}$, $\ddot{\theta}$ $\Rightarrow$ 18 incogn.
3 sòlids $\Rightarrow$ 18 eqs.

DET

x i θ només afecten **pèndol** i **suport**

Explorem sistemes que els incloguin

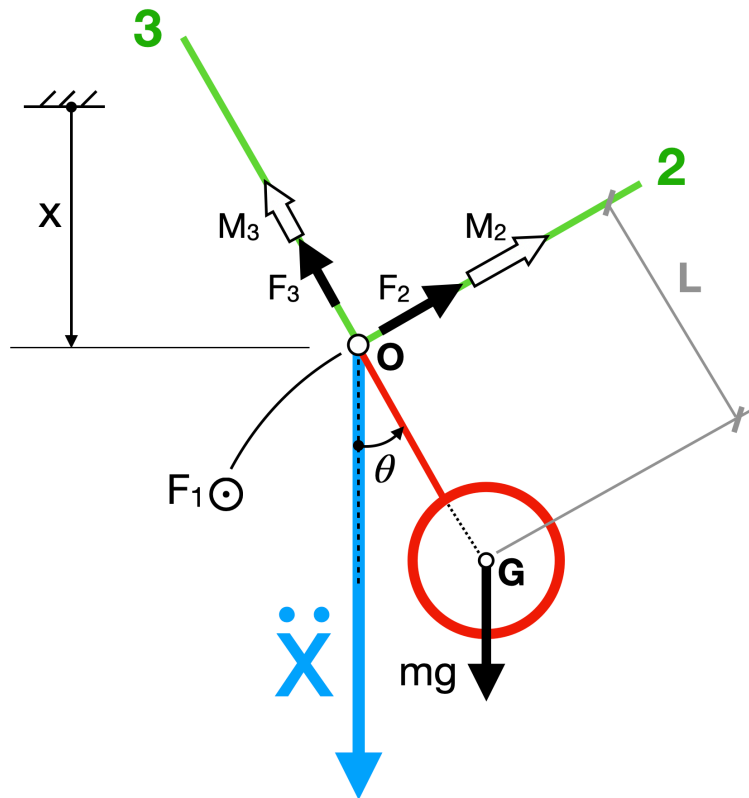
Sistema	Incògn.	Problema
Pèndol	5 ie, $\ddot{x}$, $\ddot{\theta}$	INDET
Pènd. + sup.	5 ie, $\ddot{x}$, $\ddot{\theta}$	INDET

Els altres sistemes tenen + incògnites !

Pend + Sup + Guia	8
Sup	11
Sup + guia	12
Pènd + guia	18

ER60 $\rightarrow$ Explorem aplicac^ó TQM a $\left| \begin{array}{l} \text{pènd} \\ \text{pend + sup} \end{array} \right.$ TMC

$TMC(0)]_1$ sobre sist = pèndol



Moment del pes

$$\begin{Bmatrix} 0 \\ 0 \\ -L \end{Bmatrix} \times \begin{Bmatrix} 0 \\ -mg \sin \theta \\ -mg \cos \theta \end{Bmatrix} = \begin{Bmatrix} -mgL \sin \theta \\ 0 \\ 0 \end{Bmatrix}$$

$$\sum \bar{M}_{ext}(0) - \underbrace{\overline{OG} \times m \bar{a}_T(0)}_{\text{terme complen.}} = \dot{\bar{H}}_{RTO}(0)$$

$$\left\{ \sum \bar{M}_{ext}(0) \right\}_B = \begin{Bmatrix} -mgL \sin \theta \\ M_2 \\ M_3 \end{Bmatrix} \quad (\text{I})$$

$$\left\{ \overline{OG} \times m \bar{a}_T(0) \right\}_B = \begin{Bmatrix} 0 \\ 0 \\ -L \end{Bmatrix} \times m \begin{Bmatrix} 0 \\ -\ddot{X} \sin \theta \\ -\ddot{X} \cos \theta \end{Bmatrix} = \begin{Bmatrix} -mL \ddot{X} \sin \theta \\ 0 \\ 0 \end{Bmatrix} \quad (\text{II})$$

Només pèndol té massa

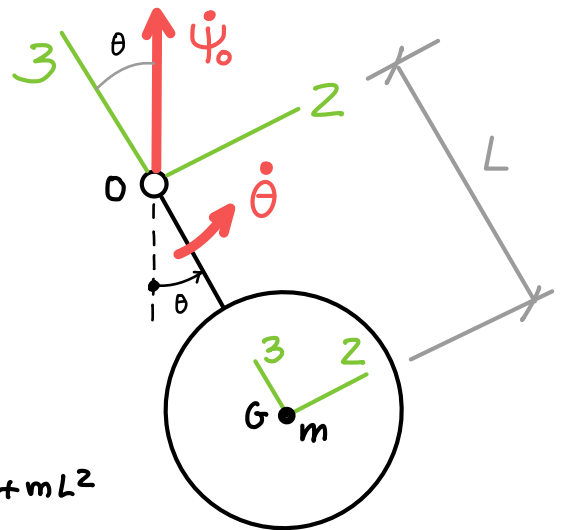
O ∈ pèndol

$$\overline{H}_{RTO}(0) = \underbrace{II(0)}_{\text{Del pèndol (anella)}} \cdot \overline{\Omega}_{RTO}^{\text{pèndol}}$$

$$[II(0)]_B \stackrel{\text{Steiner}}{=} [II(G)]_B + [II^\Theta(0)]_B$$

$$= \underbrace{\begin{bmatrix} 2I & & \\ & I & \\ & & I \end{bmatrix}}_{2I = mR^2} + \begin{bmatrix} mL^2 & & \\ & mL^2 & \\ & & 0 \end{bmatrix} =$$

$$= \begin{bmatrix} I_{11} & & \\ & I_{22} & \\ & & I_{33} \end{bmatrix} \quad \begin{aligned} I_{11} &= mR^2 + mL^2 \\ I_{22} &= m\left(\frac{R^2}{2} + L^2\right) \\ I_{33} &= \frac{mR^2}{2} \end{aligned}$$



$$\overline{\Omega}_{RTO}^{\text{pènd}} = \overline{\Omega}_{\text{sup}}^{\text{pend}} + \cancel{\overline{\Omega}_{\text{guia}}^{\text{sup}}} + \overline{\Omega}_T^{\text{guia}} = (\odot \dot{\theta}) + (\uparrow \dot{\psi}_0)$$

$$\left\{ \overline{H}_{RTO}(0) \right\}_B = \begin{bmatrix} I_{11} & & \\ & I_{22} & \\ & & I_{33} \end{bmatrix} \begin{Bmatrix} \dot{\theta} \\ \dot{\psi}_0 \sin \theta \\ \dot{\psi}_0 \cos \theta \end{Bmatrix} = \begin{Bmatrix} I_{11} \dot{\theta} \\ I_{22} \dot{\psi}_0 \sin \theta \\ I_{33} \dot{\psi}_0 \cos \theta \end{Bmatrix}$$

$$\left\{ \dot{\overline{H}}_{RTO}(0) \right\}_B = \begin{Bmatrix} I_{11} \ddot{\theta} \\ I_{22} \dot{\psi}_0 \dot{\theta} \cos \theta \\ -I_{33} \dot{\psi}_0 \dot{\theta} \sin \theta \end{Bmatrix} + \underbrace{\begin{Bmatrix} \dot{\theta} \\ \dot{\psi}_0 \sin \theta \\ \dot{\psi}_0 \cos \theta \end{Bmatrix}}_{\text{}} \times \begin{Bmatrix} I_{11} \dot{\theta} \\ I_{22} \dot{\psi}_0 \sin \theta \\ I_{33} \dot{\psi}_0 \cos \theta \end{Bmatrix} =$$

$$\text{No sempre } \overline{\Omega}_T^B = \overline{\Omega}_T^S \text{ ! } \rightarrow \overline{\Omega}_T^B = \overline{\Omega}_T^{\text{pèndol}}$$

$$\{\dot{\vec{H}}_{\text{RTO}}(\mathbf{O})\}_{\text{B}} = \begin{Bmatrix} I_{11}\ddot{\theta} + (I_{33} - I_{22})\dot{\psi}_0^2 \sin\theta \cos\theta \\ \cdot \\ \cdot \end{Bmatrix} \quad (\text{III})$$

Fent $[(\text{I}) - (\text{II}) = (\text{III})]_1 :$

$$-mgL \sin\theta + mL\ddot{x} \sin\theta = \underbrace{I_{11}}_{m(R^2+L^2)} \ddot{\theta} + \underbrace{(I_{33}-I_{22})}_{-mL^2} \dot{\psi}_0^2 \sin\theta \cos\theta$$



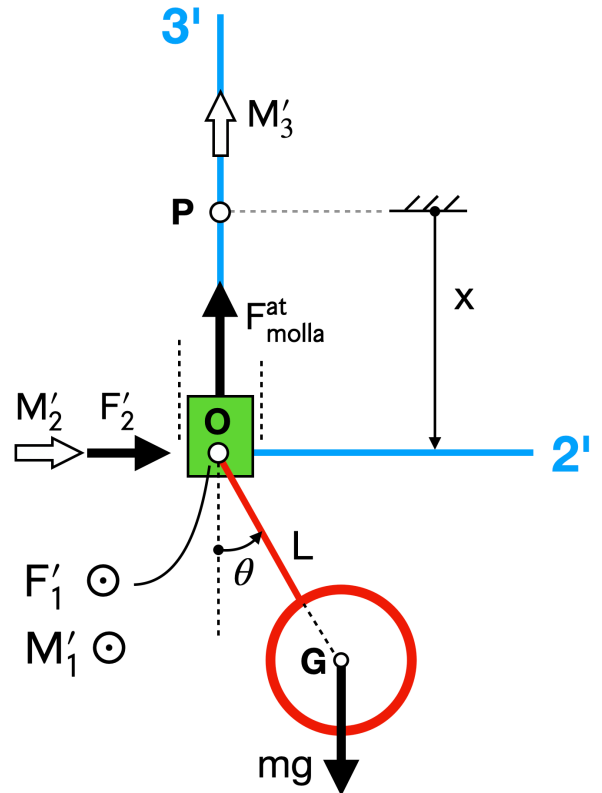
$$(R^2+L^2)\ddot{\theta} - L \sin\theta \ddot{x} = (L\dot{\psi}_0^2 \cos\theta - g)L \sin\theta$$

Eq. mov. per a θ

TQM] _{3'} sist = pendol + sup

$$\sum \bar{F}_{ext} = m \bar{a}_T(G)$$

$$\sum \bar{F}_{ext} = \begin{Bmatrix} F'_1 \\ F'_2 \\ \underbrace{mg + kx - mg}_{F^{at\ molla}} \end{Bmatrix} \quad (A)$$



$$\{\bar{p}_G\}_{B'} = \begin{Bmatrix} 0 \\ L \sin \theta \\ -x - L \cos \theta \end{Bmatrix}$$

$$\downarrow d/dt$$

$$\{\bar{v}_T(G)\}_{B'} = \begin{Bmatrix} \dot{} \\ \dot{} \\ -\dot{x} + L\dot{\theta} \sin \theta \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\psi} \end{Bmatrix} \times \begin{Bmatrix} 0 \\ L \sin \theta \\ -x - L \cos \theta \end{Bmatrix} = \begin{Bmatrix} \dot{} \\ \dot{} \\ -\dot{x} + L\dot{\theta} \sin \theta \end{Bmatrix}$$

$$\downarrow d/dt$$

$$\{\bar{a}_T(G)\}_{B'} = \begin{Bmatrix} \ddot{} \\ \ddot{} \\ -\ddot{x} + L\ddot{\theta} \sin \theta + L\dot{\theta}^2 \cos \theta \end{Bmatrix} \quad (B)$$

$$(A)]_{3'} = m \cdot (B)]_{3'} :$$

$$kx = m (-\ddot{x} + L\ddot{\theta} \sin \theta + L\dot{\theta}^2 \cos \theta)$$

$$\ddot{x} + \frac{k}{m} x - L (\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta) = 0$$

$$-(L \sin \theta) \ddot{\theta} + \ddot{x} = L \dot{\theta}^2 \cos \theta - \frac{k}{m} x$$

Eq. mov.
coord. x

Pas final

$$\begin{cases} (R^2 + L^2) \ddot{\theta} - (L \sin \theta) \ddot{x} = (L \dot{\psi}_0^2 \cos \theta - g) L \sin \theta \\ -(L \sin \theta) \ddot{\theta} + \ddot{x} = -\frac{k}{m} x + L \dot{\theta}^2 \cos \theta \end{cases}$$

↓
Aillant $\ddot{\theta}$ i $\ddot{x}$

$$\begin{cases} \ddot{\theta} = F_1(\theta, \dot{\theta}, x, \dot{x}) & \text{Eq. mov. } \theta \\ \ddot{x} = F_2(\theta, \dot{\theta}, x, \dot{x}) & \text{Eq. mov. } x \end{cases}$$

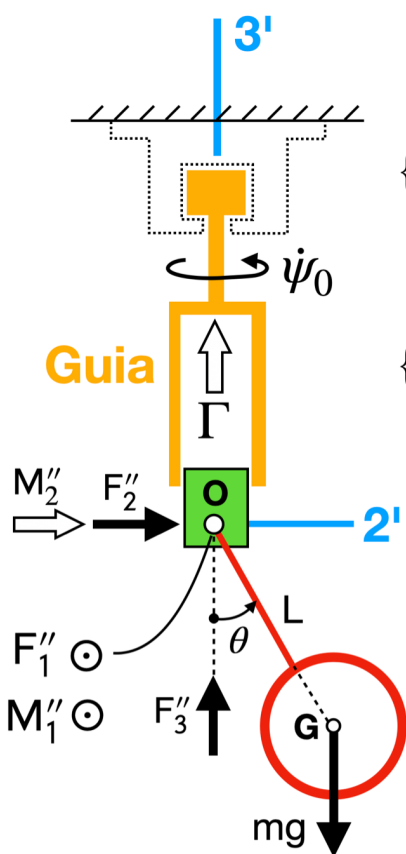
No cal
que
ho feu

Parell motor per mantenir $\dot{\psi} = \dot{\psi} = ct$

Per trobar $\Gamma \rightarrow$ Γ ha de ser interacció externa
 $\Downarrow$
 Sist. ha d'incloure la guia

Per sist = guia + pèndol + sup. Γ apareix en la comp. 3' (vertical)
 del TMC, sense i.e.

Parell motor per mantenir $\psi_0 = ct$

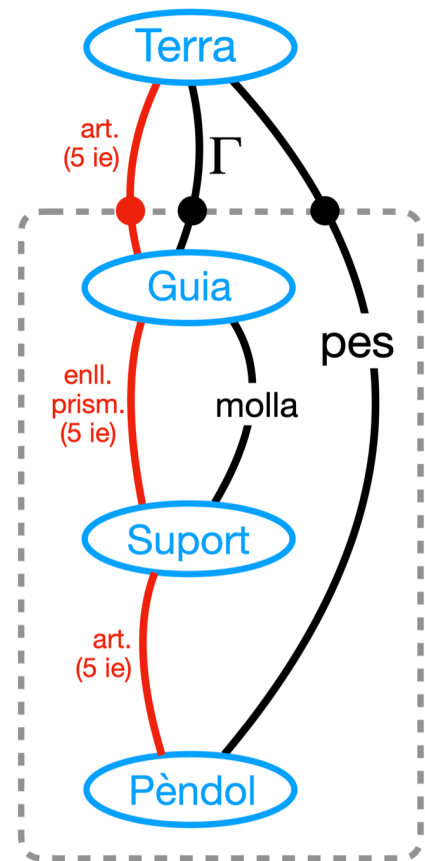


$$\{\bar{F}_{T \rightarrow \text{Guia}}\}_{B'} = \begin{Bmatrix} F_1'' \\ F_2'' \\ F_3'' \end{Bmatrix}$$

$$\{\bar{M}_{T \rightarrow \text{Guia}}(O)\}_{B'} = \begin{Bmatrix} M_1'' \\ M_2'' \\ 0 \end{Bmatrix}$$

Caract a 0 pq

- Caract. imm.
- Ja tenim $\bar{H}_{RTD}$ d'abans



$$\left\{ \sum \bar{M}_{\text{ext}}(O) \right\}_{B'} = \begin{Bmatrix} M_1'' - mgL \cos \theta \\ M_2'' \\ \boxed{\Gamma} \end{Bmatrix} \Rightarrow$$

SOL!

FULL RUTA PER Γ	Sist = Tot $TMC(O)]_{\text{vert} = 3'}$
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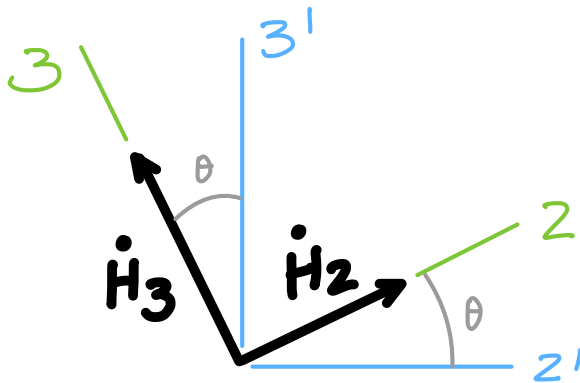
$$\underline{TMC(0)]_{3'} \quad \text{sist} = \text{Tot}}$$

$$\underbrace{\sum \bar{M}_{ext}(0)]_{3'}}_{\Gamma} - \underbrace{(\bar{OG} \times m \bar{a}_T(0))}_{\bar{O} \leftarrow (\downarrow L) \times (\downarrow m\ddot{x}) \text{ té dir } \otimes (\text{no } \uparrow)}]_{3'} = \dot{\bar{H}}_{RTO}(0)]_{3'}$$

Tenim $\dot{\bar{H}}_{RTO}(0)$ d'abans però en base B:

$$\left\{ \dot{\bar{H}}_{RTO}(\mathbf{O}) \right\}_B = \begin{Bmatrix} I_{11}\ddot{\theta} + (I_{33} - I_{22})\dot{\psi}_0^2 \sin\theta \cos\theta \\ (I_{11} + I_{22} - I_{33})\dot{\psi}_0 \dot{\theta} \cos\theta \\ (I_{22} - I_{11} - I_{33})\dot{\psi}_0 \dot{\theta} \sin\theta \end{Bmatrix} = \begin{Bmatrix} \dot{H}_1 \\ \dot{H}_2 \\ \dot{H}_3 \end{Bmatrix}$$

N'extreiem la component en dir. 3' (vert):



$$\begin{aligned} \left[\dot{\bar{H}}_{RTO}(0) \right]_{3'} &= \dot{H}_2 \sin\theta + \dot{H}_3 \cos\theta = \\ &= \underbrace{2(I_{22} - I_{33})}_{mL^2} \dot{\psi}_0 \dot{\theta} \sin\theta \cos\theta = \\ &= 2mL^2 \dot{\psi}_0 \dot{\theta} \sin\theta \cos\theta \end{aligned}$$

Per tant

$$\Gamma = 2mL^2 \dot{\psi}_0 \dot{\theta} \sin\theta \cos\theta$$